Fengyuan Dai

Google Scholar | Website | X | GitHub

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Research Interests

Protein Language Model, Protein Design, Generative AI, AI for science, RLHF.

Education

Shanghai University Sep. 2019 – Jul. 2023
Bachelor of Computer Science and Technology

ZJU & Westlake University Sep. 2023 – Jul. 2028(Expected)
Joint Ph.D. student, Computer Science

Research Profile

Ph.D. student in computer science at Zhejiang University and Westlake University, developing generative models for functional protein design. My research combines large-scale pretraining and multimodal learning to connect natural-language instructions with protein sequences. As a first author, I developed **Pinal** (*accepted by Nature*) for text-guided protein design and **DOMINO** for protein design guided by learned domain co-occurrence patterns. I also contribute to protein model infrastructure and interactive function understanding through SaprotHub and Evolla.

Selected Publications

* equal contribution

- **Programmable design of functional proteins from natural language.** *Nature*, 2026. Online Server
  Fengyuan Dai*, Shiyang You*, et al.

  Introduced Pinal, a 16-billion-parameter generative model pretrained on 1.7 billion protein–text pairs for designing proteins from natural language. Experiments confirmed the intended functions across four design tasks, including fluorescence and enzyme catalysis, without iterative experimental optimization.

- **DOMINO: Learning Domain Co-occurrence for Multidomain Protein Design.** *bioRxiv*, 2026. Online Server
  Fengyuan Dai*, Jin Su*, Qingsong Tan, Huijiao Yang, Xibin Zhou, Fajie Yuan.

  Introduced DOMINO, a framework that learns domain co-occurrence patterns in natural proteins and uses these patterns to guide domain recombination for multidomain protein design.

- **Democratizing protein language model training, sharing and collaboration.** *Nature Biotechnology*, 2025. Online Server
  Jin Su, Zhikai Li, Tianli Tao, Chenchen Han, Yan He, **Fengyuan Dai**, et al.

  Introduced SaprotHub, a platform for training, prediction and model sharing without coding. Its Google Colab server enables collaborative development of customized protein models. In this work, I trained the 1.3-billion-parameter SaProt model.

- **Decoding the Molecular Language of Proteins with Evolla.** *bioRxiv*, 2025. Online Server
  Xibin Zhou*, Chenchen Han*, . . . , **Fengyuan Dai**, et al.

  Introduced Evolla, an interactive protein-language model trained on 546 million protein–text pairs and refined through direct preference optimization. Evolla enables natural-language queries about protein function and achieves zero-shot performance comparable to state-of-the-art supervised prediction.

Awards

- National Scholarship 2026
- First Class Academic Scholarship, 2022
- Second Class Academic Scholarship, 2020, 2021

Technical Skills

Languages : Chinese (native), English(fluent).
Others : Python, PyTorch, LaTeX, Linux, Git.